AUTONOMOUS PEER-TO-PEER ENERGY SHARING AND TRADING IN DECENTRALIZED MICROGRIDS

by

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CSE4197 Engineering Project 1

Project Specification Document

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1. PROBLEM STATEMENT

Microgrids are defined as clusters of loads and microsources operating as a single controllable system [1]. Traditional centralized energy distribution systems are struggling to keep up with the growing number of distributed energy resources such as rooftop solar panels and residential battery systems. These grids were originally designed for one-way energy flow from large power plants to consumers, which makes them inefficient and inflexible when dealing with local energy surpluses or shortages. As more households start generating their own electricity, they become "prosumers" units that both produce and consume energy but the existing infrastructure offers no practical way for them to trade energy with each other directly.

This project focuses on the lack of a decentralized, autonomous software system that would allow prosumers to share and trade their surplus energy on a peer-to-peer (P2P) basis, without going through a central authority [2]. Currently, prosumers have no choice but to sell excess energy back to the main grid at low prices while buying it back at much higher rates. This imbalance reduces the economic benefit of owning renewable energy systems and slows down the wider adoption of clean energy solutions.

2. PROBLEM DESCRIPTION AND MOTIVATION

The way energy is produced and distributed is changing. Instead of relying entirely on large power plants and long transmission lines, communities are increasingly generating their own electricity through solar panels, wind turbines, and battery storage. However, the software and market infrastructure needed to support this shift has not kept pace with the hardware.

The main issue is that prosumers currently operate in isolation within a centralized system. When a household generates more energy than it needs, that surplus is either wasted or sold back to the grid at an unfavorable rate. At the same

time, a neighbor who needs extra energy has to buy it from the grid at full price. A local P2P trading system could eliminate this inefficiency by connecting buyers and sellers directly within the same community.

In Turkey, this problem is particularly relevant. Infrastructure limitations restrict how much distributed renewable energy can be absorbed by the national grid, and the existing two-tier tariff structure makes the cost gap between selling and buying energy even more noticeable. The ongoing rollout of smart meters across the country is a positive step, as it provides the real-time data exchange needed to support automated local energy trading in the future [3].

The motivation behind this project is to build a working simulation that shows how autonomous software agents can negotiate and execute energy trades within a microgrid, without any central coordination. Handling many different agents with different energy profiles in real time requires intelligent decision-making, dynamic pricing, and fast communication none of which are supported by today's centralized systems. By combining agent-based modeling, brokerless P2P communication, and a Continuous Double Auction mechanism, this project aims to demonstrate a practical and scalable approach to decentralized energy management that could help reduce costs, improve energy efficiency, and support the development of smarter, more resilient communities [4].

3. MAIN GOAL AND OBJECTIVES

The main goal of this project is to design and develop an autonomous microgrid simulation that allows residential units to share and trade energy with each other on a peer-to-peer basis, without any central control or coordination. To achieve this, the following objectives have been defined:

- Objective 1: To model and implement autonomous software agents representing prosumers and consumers, each with their own energy profile, storage capacity, and decision-making logic using the Mesa framework.

- Objective 2: To design and implement a decentralized, brokerless P2P communication layer using ZeroMQ, enabling agents to exchange energy status information and negotiate trades in real time.

- Objective 3: To develop a market clearing mechanism based on a Continuous Double Auction that automatically matches energy buyers and sellers based on price and availability.

- Objective 4: To build an event-driven simulation environment using SimPy that processes time-series data of solar generation and household consumption, and models battery storage behavior over time.

- Objective 5: To evaluate the overall system performance using metrics such as transaction success rate, message latency, and reduction in macro-grid energy dependency.

4. RELATED WORK

The development of an autonomous P2P energy trading system draws on several areas of existing research, including microgrid standards, market mechanisms, simulation tools, and communication protocols.

4.1 Microgrid Standardization and Control

The IEEE 2030.7 and 2030.8 standards define the core operational framework for microgrids, covering functions such as dispatch, load balancing, and transitions between grid-connected and islanded modes [5]. These standards typically assume a centralized energy management system. However, more recent research has explored distributed control approaches in which nodes coordinate through local communication rather than relying on a central controller. This shift toward decentralized architectures is directly relevant to our project, as we aim to remove the central authority from the energy management process entirely.

4.2 Market Clearing and Auction Mechanisms

Vieira and Zhang [2] compare two auction-based approaches for decentralized energy markets: Continuous Double Auctions (CDA) and Uniform-Price Double-Sided Auctions. Their work highlights that CDAs are well suited for real-time energy trading because they allow continuous matching of buyers and sellers based on price and time priority. Game-theoretic models are also explored to capture the competitive behavior of sellers and the risk preferences of buyers. Based on this

literature, we have chosen the CDA as the market clearing mechanism for our simulation. The proposed decentralized microgrid architecture, which utilizes this CDA mechanism for peer-to-peer (P2P) energy trading, is illustrated in Figure 1.

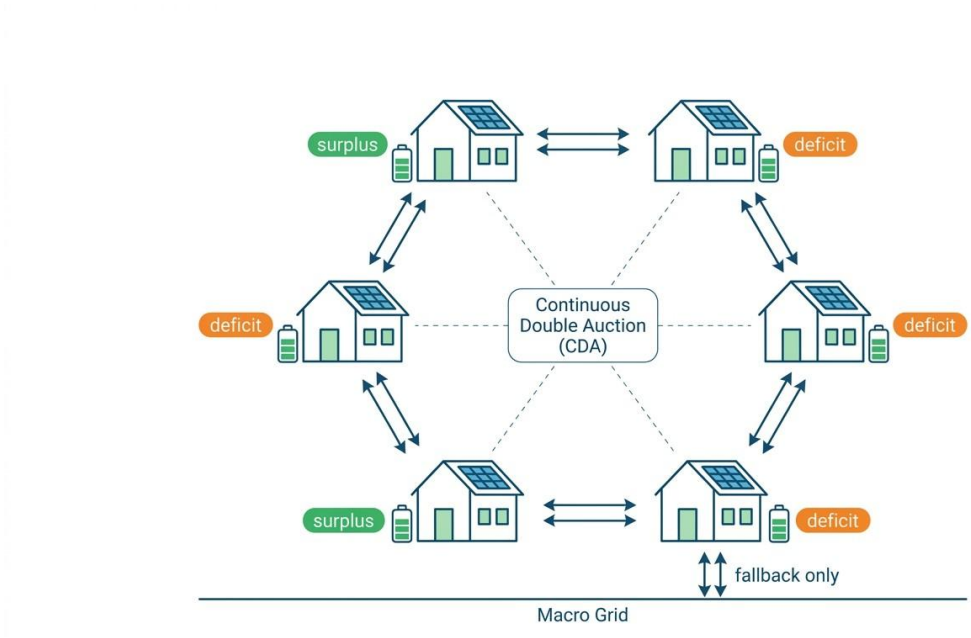


Figure 1: Decentralized Microgrid Architecture for P2P Trading via Continuous Double Auction (CDA)

4.3 Agent-Based Simulation Frameworks

Agent-Based Modeling is a widely used approach for studying the behavior of decentralized systems with many interacting agents. Python-based frameworks like Mesa 3.0 have become popular for this purpose due to their flexibility and compatibility with data analysis libraries [6]. For modeling physical events with precise timing such as battery charge and discharge cycles researchers often combine ABM with discrete-event simulators like SimPy. Our project follows this same combination, using Mesa for agent logic and SimPy for event-driven energy flow simulation [7].

4.4 Decentralized Communication Protocols

Reliable message exchange between agents is essential for any P2P trading system. MQTT is a common choice in IoT applications, but it depends on a central

broker, which introduces a single point of failure. ZeroMQ offers an alternative by providing socket-based, brokerless communication with lower latency in scenarios with moderate to high message traffic. For a system where agents must negotiate and respond in real time without any central coordination, ZeroMQ is the more appropriate choice, and it is the communication library we use in this project [8].

4.5 Turkish Regulatory and Economic Context

Turkey's residential energy market operates under a two-tier tariff system, where the rate paid for selling surplus energy back to the grid is significantly lower than the rate paid for purchasing it [9] [10]. This gap makes local P2P trading economically attractive for prosumers. Additionally, the ongoing national rollout of smart meters provides the real-time metering infrastructure that is a basic technical requirement for any automated local energy trading system. These conditions make Turkey a relevant and practical context for validating the approach developed in this project.

5. SCOPE

This project is limited to the design and development of a software-based simulation environment for autonomous P2P energy sharing and trading within a decentralized microgrid. No physical hardware, power electronics, or real grid infrastructure is involved.

5.1 Included in The Project Scope

- Modeling of autonomous prosumer and consumer agents with individual energy profiles, storage states, and decision-making logic using Python and Mesa 3.0.

- Implementation of a brokerless P2P communication layer using ZeroMQ, supporting real-time message exchange between agents.

- Development of a Continuous Double Auction (CDA) market clearing mechanism to match energy buyers and sellers dynamically.

- Simulation of energy flow and battery storage behavior over time using SimPy.

- Performance evaluation based on simulated time-series data, measuring transaction success rate, communication latency, and macro-grid dependency.

5.2 Excluded from The Project Scope

- Physical deployment of microgrid hardware, solar panels, or battery systems.

- Implementation of blockchain-based consensus mechanisms.

- Modeling of low-level electrical phenomena such as AC power flow, frequency regulation, or reactive power.

- A production-ready user interface or public-facing web application.

5.3 Constraints

- **Budget:** The project is executed using institutional research resources and high-performance open-source libraries. Budgetary efficiency is achieved by utilizing an entirely open-source software stack (Python, Mesa, ZeroMQ), eliminating the need for proprietary licensing fees.

- **Computation:** Simulation complexity is constrained by the processing power of standard student laptops .

- **Data:** Due to privacy regulations regarding smart meter telemetry, the project relies on high-resolution synthetic datasets and anonymized historical profiles adjusted for the Istanbul metropolitan region.

5.4 Assumptions

- The entire software stack is open-source (Python, Mesa, ZeroMQ, SimPy), eliminating any licensing costs.

- Simulation scale is limited by the processing capacity of standard development machines.

- Real smart meter data is not available due to privacy regulations; the project will use synthetic datasets and publicly available energy consumption profiles instead.

6. METHODOLOGY AND TECHNICAL APPROACH

6.1 Multi-Agent System (MAS) with Mesa

The core of the simulation is built using Mesa 3.0, a Python-based agent-based modeling framework. Each residential unit in the microgrid is represented as an autonomous agent with its own energy profile, including solar generation capacity, household consumption pattern, and battery storage state. At each simulation step, agents evaluate their current energy balance, the difference between generation and consumption and decide whether to place a sell offer, a buy bid, or take no action. Agents with surplus energy submit offers to the market, while agents in deficit submit bids. The Mesa scheduler manages the order and timing of agent activations throughout the simulation [6].

6.2 Communication Layer with ZeroMQ

Agent-to-agent communication is handled through ZeroMQ, a brokerless messaging library that enables direct socket-based communication without a central message broker. Two messaging patterns are used: the REQ-REP pattern for direct one-to-one negotiations between agents, and the PUB-SUB pattern for broadcasting energy status updates to all agents in the network. This setup ensures that no single point of failure exists in the communication infrastructure and that agents can exchange information in real time without depending on a central coordinator [8].

6.3 Market Mechanism: Continuous Double Auction (CDA)

The market clearing logic is based on a Continuous Double Auction mechanism. Agents with surplus energy act as sellers and submit offers specifying the quantity available and the minimum acceptable price. Agents with a deficit act as buyers and submit bids with the quantity needed and the maximum price they are willing to pay. The CDA module processes these submissions continuously and executes a trade

whenever a buyer's bid price meets or exceeds a seller's offer price. Matched transactions are recorded along with the traded quantity and agreed price. This mechanism was chosen because it supports real-time, decentralized price discovery without requiring a central pricing authority [11].

6.4 Discrete-Event Simulation with SimPy

SimPy is used alongside Mesa to model time-sensitive physical processes that are difficult to capture in a step-based agent model. Specifically, SimPy handles battery charge and discharge dynamics, including storage limits and degradation over time, as well as the timing of energy requests and delivery events. The combination of Mesa for agent behavior and SimPy for event-driven physical processes allows the simulation to more accurately reflect how a real microgrid would operate over an extended period.

The overall workflow of the methodology is summarized below:

- Agents are initialized with individual energy profiles and battery states.
- At each time step, each agent calculates its current energy balance.
- Agents with surplus submit sell offers; agents with deficit submit buy bids via ZeroMQ.
- The CDA module matches compatible offers and bids and records completed trades.
- Unmatched energy deficit falls back to the macro-grid.
- SimPy updates battery states and physical constraints after each trading cycle.[9]
- The process repeats across the full simulation period, and performance metrics are collected.

7. PROFESSIONAL CONSIDERATIONS

7.1 Methodological Considerations / Engineering Standards

The project follows standard software engineering practices throughout development. Version control is managed using Git and GitHub for collaborative development and change tracking. Google Drive and Google Docs are used for shared documentation. The tools and libraries used are summarized in the table below.

Category	**Tools**
Programming Language	Python 3.x
Agent-Based Modeling	Mesa 3.0
Communication	ZeroMQ
Discrete-Event Simulation	SimPy
Data Handling	Pandas, NumPy
Visualization	Matplotlib, Plotly
Version Control	Git, GitHub
Documentation	Google Drive, Google Docs

Table 1: Software Ecosystem for the P2P Energy Trading Model

7.2 Economical

The project uses an entirely open-source software stack, which eliminates licensing costs. No physical hardware is required beyond standard development laptops. The system being modeled has direct economic relevance: by enabling local energy trading, it can reduce household electricity costs by allowing prosumers to sell surplus energy at better rates than the macro-grid feed-in tariff and allowing buyers to purchase at lower prices than the retail rate.

7.3 Environmental

The project itself has no physical environmental impact. The system it models, however, is directly aligned with environmental goals. By maximizing the local use of renewable energy and reducing the amount of energy drawn from the macro-grid, P2P microgrid trading lowers the overall carbon footprint of a residential community.

7.4 Health and Safety

This project involves only software development and simulation. There are no physical components, electrical systems, or laboratory equipment involved, so there are no health or safety concerns associated with this project.

7.5 Sustainability

The open-source nature of the entire software stack ensures the project remains accessible and maintainable beyond its initial development. The system it models directly supports long-term sustainability by promoting the efficient use of locally generated renewable energy and reducing community dependence on centralized, fossil-fuel-based power.

7.6 Social

P2P energy trading can help distribute the benefits of renewable energy more equitably across a community. Households that cannot install solar panels due to cost or physical limitations can still benefit from clean, locally traded energy at competitive prices. This project provides a simulation framework that can be used to

study and demonstrate these social benefits, which may inform future community energy policies.

7.7 Legal

All software used in this project is open-source and carries no restrictive licensing obligations. The simulation relies exclusively on synthetic or publicly available datasets, so no personal data regulations apply. If the system were to be deployed in a real-world setting in the future, compliance with relevant national energy market regulations such as Turkey's Balancing and Settlement Regulation would need to be addressed. At the current stage of research and simulation, no legal concerns apply.

8. MANAGEMENT PLAN

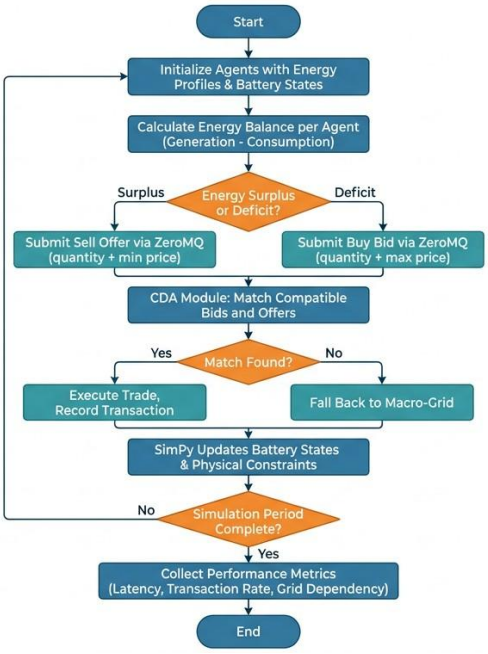


Figure 2: Workflow of the Proposed P2P Energy Trading Simulation

8.1 Description of Task Phases

Phase 1: Literature Review and Requirement Analysis

This phase involves a deep dive into existing P2P energy trading literature, IEEE 2030.7/8 standards, and the current EPDK regulatory landscape in Turkey to

define the system's technical requirements. We will establish the theoretical framework for prosumer behavior models and the specific data structures needed for the simulation.

Phase 2: Modeling of Agent Logic and Energy Profiles in Mesa

Using the Mesa 3.0 framework, we will implement the state machines for autonomous software agents, defining their individual energy profiles and battery storage logic. Agents will be programmed to calculate net energy and determine optimal trading strategies based on their State-of-Charge (SoC).

Phase 3: Implementation of ZeroMQ P2P Communication Infrastructure

We will develop a decentralized, brokerless communication layer using ZeroMQ to facilitate real-time, asynchronous messaging between agents. This includes implementing PUB-SUB patterns for market broadcasting and REQ-REP patterns for direct trade negotiations.

Phase 4: Development of the CDA Market Clearing Algorithm

A Continuous Double Auction (CDA) mechanism will be developed to match buyer bids and seller asks dynamically based on price and time priority. The algorithm will handle clearing price calculations and include automated fallback procedures to the macro-grid when local matches are unavailable.

Phase 5: Integration of Co-simulation (Mesa-SimPy) and Performance Testing

This phase integrates Mesa’s economic agent logic with SimPy’s discrete-event simulation to ensure the virtual grid respects physical timing and temporal constraints. The system will be stress-tested to measure message latency, transaction success rates, and overall energy cost reductions.

Phase 6: Data Analysis and Final Documentation

The final phase focuses on interpreting the simulation results to evaluate the project's overall performance and scientific impact. We will complete the final project report, consolidate all technical documentation, and archive the source code to ensure the project's sustainability and future extensibility.

8.2 Division of Responsibilities

Member	Primary Responsibility
Erol Kahyaoğlu	MAS Architecture, Agent Classes, Spatial Topology
Kutluay Talay	ZeroMQ Communication, Data Pipelines, Messaging Patterns
Yusuf Samet Pek	Economics, CDA Algorithm, Financial State Management

Table 2: Task Distribution and Team Member Responsibilities

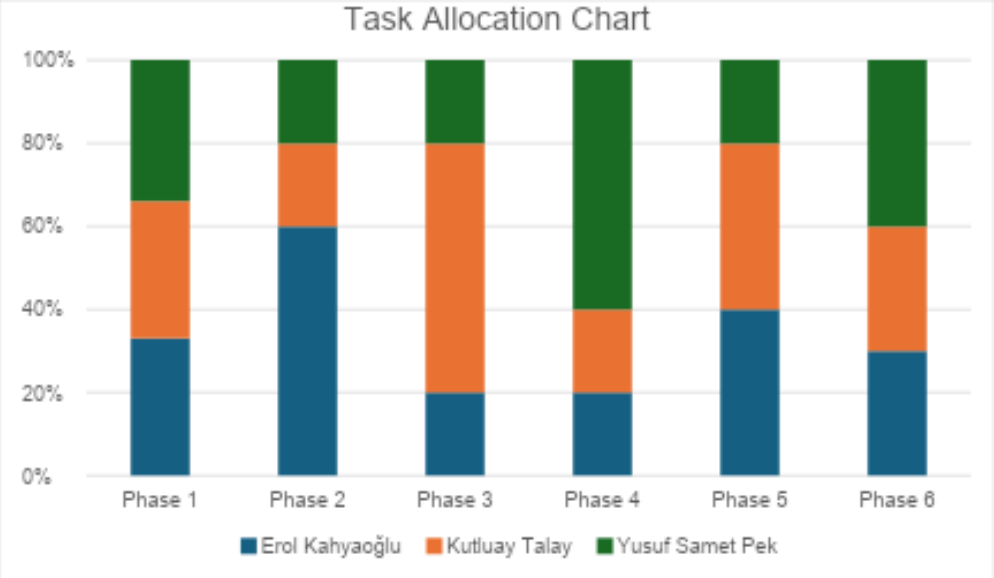


Figure 3: Task Allocation Chart

8.3 Time Line with Milestones

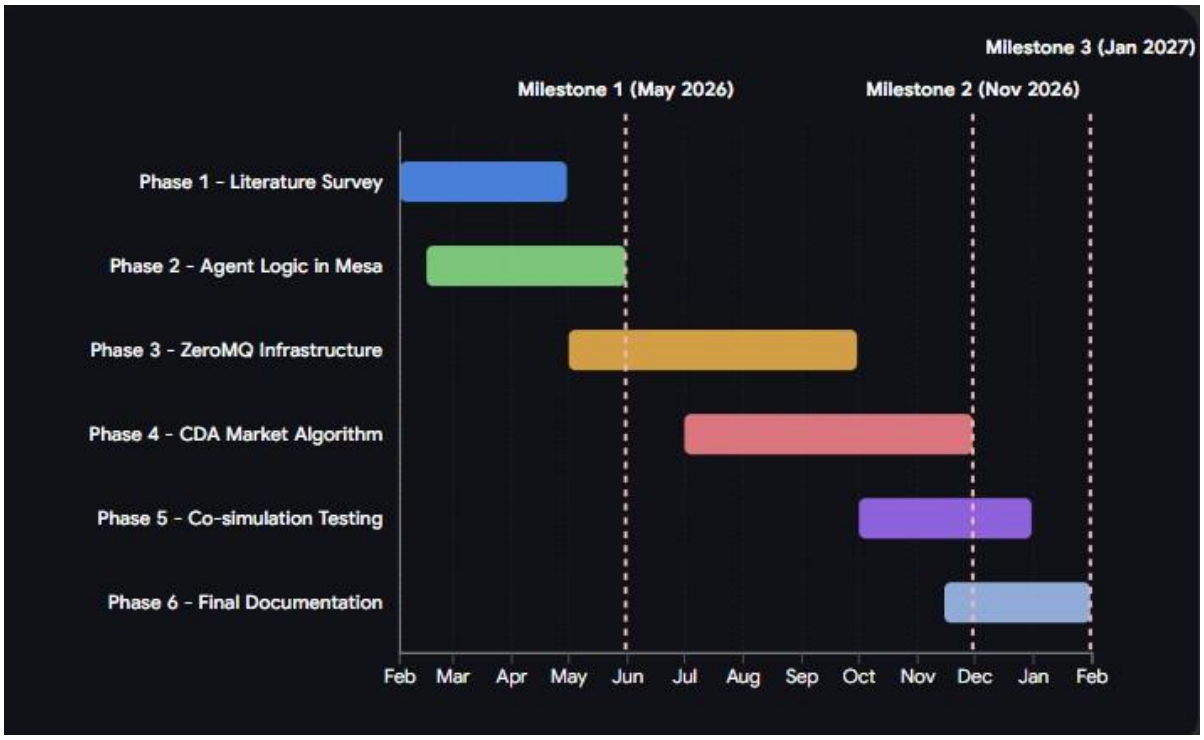


Figure 4: Timeline with Milestones

- **Milestone 1 (May 2026):** Basic MAS environment with local inference/logic.
- **Milestone 2 (Nov 2026):** Fully functional P2P communication and trading matching.
- **Milestone 3 (Jan 2027):** Final simulation results and performance evaluation.

9. SUCCESS FACTORS AND RISK MANAGEMENT

9.1 Measurability / Measuring Success

The project will be considered successful if the following criteria, each tied to a specific project objective, are met:

Objective 1 — Agent Modeling (Mesa):

Success Factor: Each agent correctly maintains its own energy balance, storage state, and decision logic throughout the simulation without errors across varying energy scenarios.

Objective 2 — P2P Communication (ZeroMQ):

Success Factor: All agents exchange energy status messages and trade bids through ZeroMQ without message loss or blocking under the simulated load.

Objective 3 — Market Clearing (CDA):

Success Factor: The CDA mechanism successfully matches a majority of compatible buy and sell offers each cycle, with all transactions correctly logged with price and quantity values.

Objective 4 — Simulation Environment (SimPy):

Success Factor: The simulation runs stably over the full period using time-series data, with no physical constraint violations and consistent synchronization between Mesa and SimPy.

Objective 5 — Performance Evaluation:

Success Factor: The simulation produces measurable results showing a reduction in macro-grid energy dependency compared to a no-P2P baseline, with transaction rate, latency, and grid dependency metrics reported for at least one complete run.

9.2 Risk Management

Risk	Impact	Mitigation Strategy
Mesa-ZeroMQ Synchronization: Potential timing conflicts between the agent scheduler and the asynchronous communication layer.	High	Developing a small-scale (3-node) prototype to validate message handling before full-scale simulation.
Computational Bottlenecks: High CPU/RAM usage on standard laptops when simulating many agents.	Medium	Utilizing **vectorized operations** with NumPy and optimizing the Mesa scheduler logic.
Data Scarcity: Lack of high-resolution real-time smart meter data due to privacy regulations.	Low	Using validated open-source synthetic datasets and scaling them to Istanbul's solar irradiation profiles.
Convergence Issues: CDA market failing to clear during high volatility periods.	Medium	Implementing automated fallback mechanisms to the macro-grid to ensure energy stability.

Table 3: Risk Management and Mitigration table

10. BENEFITS AND IMPACT OF THE PROJECT

Scientific impact: This project contributes to the field of transactive energy systems by providing a high-fidelity, open-source co-simulation testbed. The

integration of Agent-Based Modeling (ABM) with brokerless P2P communication (ZeroMQ) offers a novel framework for studying emergent market behaviors in decentralized microgrids. The findings are intended to support further academic research into autonomous energy negotiation strategies.

Economic Impact: The proposed software prototype optimizes local consumption, directly reducing household electricity bills. This is particularly valuable under Turkey's **EPDK two-tier tariff system**, where the gap between buying and selling prices is significant.

Social & Environmental Impact: By promoting the efficient use of rooftop solar and local storage, the project supports the transition to green energy. It fosters energy equity by allowing residents without solar installations to purchase cheaper, renewable energy from their neighbors.

National Energy Resilience: Decentralized microgrids enhance energy security by enabling localized sharing during macro-grid outages or peak-shaving periods, reducing the overall load on the national infrastructure.

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